TutorConnect NG

MINIMUM VIABLE PRODUCT
PRODUCT REPORT & FUTURE ROADMAP

Product	Document	Owner	Class
TutorConnect NG — Verified Tutor Marketplace for Nigeria	MVP Product Report · Version 2.0	Team TutorConnect	Confidential

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1 | Executive Summary

TutorConnect NG is a full-stack education technology (EdTech) marketplace that connects Nigerian students and parents with **verified** local and online tutors. The minimum viable product — a production-grade web application built on Next.js 16, React 19, TypeScript and Neon Serverless Postgres — delivers the complete end-to-end journey: discovery of tutors, intelligent filtering by subject, location, budget and curriculum, verified tutor profiles, a booking flow with live pricing in Naira, role-based dashboards for four user personas, and **private in-platform chat** between students or parents and tutors.

The MVP was engineered to answer one strategic question: *can we deliver a trust-first tutoring marketplace for Nigeria — where every publicly listed tutor has been human-vetted for identity and credentials — at startup speed, without compromising data integrity?* The delivered product answers yes: it enforces an admin-gated verification pipeline, never fabricates marketplace data, and keeps both sides of the market engaged through dashboards and private messaging.

2 | Problem & Opportunity

The problem	The opportunity
Nigerian families depend on private tutoring for WAEC, JAMB, NECO, Post-UTME and international curricula (British Cambridge, American), yet discovery happens through word-of-mouth. Pricing is opaque, tutor quality is unverifiable, and parents have no way to confirm the identity or qualifications of the person teaching their child — online or in person. Tutors, in turn, lack a professional channel to reach students, manage availability and grow income.	A single trusted marketplace where every tutor is verified (government ID + academic credential review), searchable by subject, state/LGA, budget and curriculum, instantly bookable with transparent Naira pricing, and reachable through private on-platform chat — without exposing personal phone numbers. Trust, made the product's core differentiator.

3 | MVP Scope — Delivered Capabilities

3.1 Discovery & Live Matching Engine

- Homepage and tutor directory with a **live filter & matching engine**: subject/skill, state and locality (Lagos — Lekki, Ikeja, Yaba; Abuja — Maitama, Gwarinpa; Port Harcourt; Ibadan; Online/Remote), a Naira budget slider, weekday/weekend availability, and curriculum (Nigerian National, British Cambridge, American).
- Eight-subject catalogue with live tutor-count badges: Mathematics & Further Maths, English & Literature, Sciences, Full-Stack Web Development & Python, Generative AI & Data Skills, UI/UX & Digital Design, Exam Prep (NECO, Post-UTME, SAT, IELTS) and Foundational Literacy & Numeracy.

3.2 Verified Tutor Profiles — Trust by Design

- Public profile pages with verified-ID badges, credentials, headline and bio, hourly rate in Naira, years of experience, star ratings with written reviews, and an interactive weekly availability calendar.
- **Strict visibility rule:** every registration starts as *pending*; a tutor becomes publicly visible only after an administrator reviews their ID-card and degree uploads and approves them. Only verified profiles are served by the tutor API — and no tutor, booking or review data is ever fabricated or seeded; the interface shows explicit empty states instead.

3.3 Booking Flow & Session Lifecycle

- Booking modal capturing grade level, subject, date & time slot, session mode — online (Google Meet/Zoom) or in-person — and notes, with a live total-price estimate in Naira.
- Full booking lifecycle: *pending → accepted / rejected → completed / cancelled*, with reviews bound 1:1 to completed bookings and tutor rating averages maintained automatically by database triggers.

4 | In-Platform Private Chat — Core MVP Capability

Private one-to-one messaging is built into the heart of the product and is available from **all four dashboards** (student, parent, tutor and admin). It lets families question tutors about teaching approach, curricula, scheduling and pricing *before* booking, and keeps ongoing coordination on-platform — without exposing personal phone numbers or third-party apps.

Capability	Delivered behaviour
Private threads	Conversations are strictly between the signed-in user and one other party; each chat panel carries the trust indicator *"Private — only the two of you can see these messages."*
Security & identity	JWT-enforced server-side — users chat strictly as their own account (no impersonation); recipients are validated against real registered accounts; self-messaging is rejected; messages are truncated at 2,000 characters and persisted to Postgres with thread- and participant-level indexes.
Contact discovery	A conversation list with per-partner threads plus a "Start a new chat" picker that lists only real, registered counterparties (verified tutors for students/parents; registered users for admins). No mock or demo personas are ever presented.
Delivery & UX	Threads auto-refresh on a 5-second poll; smart auto-scroll fires only for genuinely new messages; send-on-Enter, timestamps, unread-friendly conversation ordering and a fully responsive mobile/desktop layout.

5 | Role-Based Dashboards — Four Personas

Persona	Dashboard capabilities
Student	Overview, My Bookings, upcoming sessions, saved tutors, review submission, profile management and private chat with tutors.
Parent	Child profiles (name, age, educational level, tutoring budget, preferred learning mode), bookings, total-spend tracking, reviews, account management and private chat with tutors.
Tutor	Accept/decline booking requests, weekly availability management, pending/upcoming/completed session counters, earnings tracking, credential-document upload for verification and private chat.
Admin	Platform KPIs (total tutors, total bookings, pending verifications, platform revenue), user management (approve / reject / suspend / delete), credential verification queue with document viewer, dispute and metrics monitoring, and private chat with any user.

6 | Technology & Architecture

Layer	Implementation
Frontend	Next.js 16 (App Router) · React 19 · TypeScript 5 · Tailwind CSS v4 design system (navy/indigo, emerald, amber) · Framer Motion animation · lucide-react icons
Backend	Next.js API routes on Node.js runtime — tutors, bookings, reviews, messages (private chat) and authentication endpoints
Database	Neon Serverless Postgres — users, tutor profiles, subjects, availability, bookings, reviews, messages, verification documents; enums, indexes, rating-aggregation triggers and an idempotent migration system
Auth & security	JWT sessions with bcrypt password hashing; account lifecycle states (pending / approved / rejected / suspended); admin-gated tutor verification

Hosting	Vercel serverless deployment with graceful degradation — a database-less demo falls back to in-memory auth while marketplace screens show honest empty states (never fabricated data)

What the MVP validates: (i) a trust-first supply side through admin-gated verification; (ii) the demand-side conversion funnel — discover → filter → view profile → book; (iii) two-sided retention surfaces via dashboards and private chat; and (iv) the feasibility of a serverless Next.js + Postgres architecture for a two-sided marketplace.

7 | Deliberately Out of Scope in the MVP

- **Payments & escrow** — bookings capture agreed pricing; no money movement, payouts or refunds yet.
- **Real-time transport** — chat refreshes on a 5-second poll rather than WebSockets/push.
- **Native video classroom** — online sessions reference Google Meet/Zoom; no in-app video or whiteboard.
- **Notifications** — no email/SMS/push alerts for bookings, reminders or new messages.
- **Mobile apps** — responsive web only; no PWA install or native iOS/Android clients.
- **Hardened route guarding** — dashboards are intentionally left accessible to reviewers/stakeholders for demo purposes; JWT middleware enforcement is deferred to production hardening.

8 | Future Scope — A 30-Day Enhancement Roadmap

Given one further month of development, investment would follow six priorities, sequenced to convert the MVP from a trusted discovery-and-booking marketplace into a **monetised, real-time, full-lifecycle learning platform** ready for a Lagos–Abuja pilot launch.

PRIORITY 1 · WK 1 — **Monetisation — Payments & Payouts**

- Integrate **Paystack and Flutterwave**: pay-at-booking with escrow-style holding until session completion, tutor payout ledger, refunds, and a platform commission engine — turning bookings into revenue.
- Session packages and subscriptions (e.g., 10-session bundles) and discounted trial lessons to lift conversion and lifetime value.

PRIORITY 2 · WK 1–2 — **Real-Time Messaging Upgrade**

- Replace polling with **WebSockets / SSE** (Pusher, Socket.IO or Postgres LISTEN/NOTIFY): instant delivery, read receipts, typing indicators, unread badges.
- Push/email notification for new messages; image & document sharing (homework, past questions) with content moderation and spam controls.

PRIORITY 3 · WK 2–3 — **Virtual Classroom & Notifications**

- In-app video sessions via WebRTC (Daily/Jitsi) or automated Zoom/Meet link generation tied to each accepted booking — one-tap “Join session”, calendar sync, session reminders.
- Transactional notification service (email + WhatsApp/SMS via Twilio or Termii) for booking status changes, session reminders and chat activity.

PRIORITY 4 · WK 3 — **Learning Experience & Parent Insights**

- Post-session artefacts: tutor session notes, homework and attendance captured per booking.
- Parent progress reports per child — the retention and word-of-mouth engine for family accounts.

PRIORITY 5 · WK 3–4 — **Trust, Safety & Compliance**

- Production hardening: Next.js Middleware JWT/role guards on every dashboard, RBAC, rate limiting and admin audit logs.
- NDPR/GDPR-aligned data practice: encryption of chat at rest, retention policies, consent management and data export/deletion flows; KYC assistance for tutor verification (NIN/BVN check APIs).

PRIORITY 6 · WK 4 **Quality, Reliability & Growth**

- CI/CD with automated testing — Playwright end-to-end coverage of booking and chat flows, unit coverage of API routes — plus error monitoring (Sentry) and product analytics for the search → profile → booking funnel.
- Intelligent matching: ranking tutors by learning goals, budget, availability and review sentiment; installable PWA (offline shell, home-screen install) as the bridge to React Native mobile apps.

How success in the next 30 days will be measured

Metric	MVP baseline	30-day target
Booking conversion	Lifecycle tracked end-to-end	60%+ of accepted sessions reach completed
Chat engagement	Poll-based delivery across 4 dashboards	40%+ of bookings preceded by a private chat thread
Gross booking value	Not captured — no payments in MVP	First Naira revenue via Paystack/Flutterwave, with commission
Verified supply	Admin-gated verification queue	+25% verified tutors across Lagos & Abuja
Reliability	Manual QA	CI/CD with Playwright E2E; zero critical defects at pilot

Bottom line: within 30 additional days, TutorConnect NG moves from a trust-first discovery and booking MVP to a monetised, real-time learning platform — with payments, live messaging, an integrated classroom, compliance-grade security and measurable growth tooling in place.

Recommended immediate actions

- Approve the 30-day enhancement cycle, beginning with **Priority 1 (payments)** and **Priority 2 (real-time messaging)** — the two features that convert engagement into revenue and retention.
- Commission an NDPR data-protection review in parallel with Priority 5, so compliance lands before the pilot launch rather than after it.
- Freeze the MVP feature set as the production baseline; all future work is tracked against the six priorities above.

Document control	
Version	2.0
Prepared by	Team TutorConnect — Uchekwe Uchechukwu Hephzibah, Chioma Helen Ezeufonna, Bodman Ebuka Precious, Uzochukwu Mary Cynthia Enyemakachukwu, Pius Kaosisochukwu Amanie
Status	Approved for stakeholder and investor review
Next review	Upon completion of the 30-day enhancement cycle

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